Calango

User Guide

Visual atomistic modeling, simulation and analysis

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About this guide

Calango is a desktop application for building, visualizing, simulating and analysing atomic structures. It pairs a C++20 / OpenGL core — fast enough to orbit tens of thousands of atoms in real time — with an embedded CPython interpreter running the Atomic Simulation Environment (ASE), so every structure you see on screen can be handed to a real calculator without leaving the program.

This guide is written for the person using Calango, not the person building it. It walks through the workspace, then covers each menu in the order you are likely to need it: getting data in, building and editing structures, making them look right, running simulations locally or on a cluster, and analysing the results. Chapter 13 is a reference you can jump straight to — keyboard shortcuts, the complete menu map, file formats and troubleshooting.

Conventions

Analysis ⟩ **Raman Modes** A menu path. Each ⟩ step is one level deeper.

Compute A button, field, checkbox or tab label exactly as it appears in the interface.

Ctrl+**R** A keyboard shortcut. On macOS, Ctrl means Cmd wherever the shortcut is a standard editing action (open, save, quit, undo); single-letter viewport shortcuts have no modifier on any platform.

`run.py` A file, directory or path.

Three kinds of callout appear throughout:

> **Note**
> Background you do not strictly need, but that explains why something behaves the way it does.

> **Tip**
> A shortcut or working habit that saves time.

> **Requires**
> An external dependency — a Python package, a solver binary, a network connection — that the feature cannot work without.

Version

This guide documents Calango **26.7.26**. The version you are running is shown in **Help ⟩ About Calango**, together with the Python and ASE versions actually in use — worth checking first whenever something behaves unexpectedly.

Contents

CHAPTER 1

Getting Started

1.1 Installing Calango

Calango ships as a native installer for each desktop platform:

macOS A drag-and-drop `.dmg`. Mount it and drag `calango.app` onto the `Applications` shortcut.

Linux A Debian package. Install it with `apt` so dependencies resolve automatically:

```
sudo apt install ./calango_26.7.26_amd64.deb
```

The package registers a desktop launcher, an application icon, and a MIME type for `.calproj` files, so double-clicking a project in your file manager opens the workspace.

Building from source is covered in the companion *Calango Packaging Guide* (`docs/packaging.pdf`), which also documents how the installers are produced.

1.2 The Python environment

Calango embeds a CPython interpreter and drives ASE through it. Almost everything that touches chemistry — reading a CIF, building a slab, running a calculation — goes through that interpreter, so pointing Calango at a Python that has ASE installed is the single most important piece of setup.

1.2.1 How the interpreter is chosen

An embedded interpreter does not inherit an activated virtual environment the way a shell does, so Calango resolves one explicitly, in this order:

1. `$CALANGO_PYTHON` — an explicit path to a Python executable. Highest priority; use this to override everything else.
2. `$VIRTUAL_ENV` — an activated virtual environment, if Calango was launched from that shell.
3. A Python bundled inside the installed application (`Contents/Resources/python` on macOS, `<prefix>/lib/calango/python` on Linux), when the installer was built with one.
4. The interpreter the build was configured against.

Requires

Calango needs `ase` and `numpy` at minimum. Individual features add their own requirements: `spglib` (symmetry, Raman modes), `paramiko` (remote access), `pillow` / `imageio` / `imageio-ffmpeg` (animation export), `mace-torch` (ML potentials), `gpaw` (DFT bands and PDOS), `icet` (optional, improves SQS generation). A complete environment:

```
python3 -m venv .venv
.venv/bin/pip install ase numpy spglib paramiko pillow imageio \
                      imageio-ffmpeg
```

1.2.2 Checking the environment

Before opening any structure, confirm the interpreter is the one you expect:

```
calango --probe-python
```

This runs headlessly and prints the resolved interpreter, the Python version and the ASE version, exiting with status 0 only when ASE imports successfully. It is the first thing to run when structure loading fails.

If ASE cannot be imported at launch, Calango still starts — so you can adjust settings — but structure I/O and job features are disabled, and a warning explains how to point at a working interpreter.

Tip

Simulation jobs do not have to use the embedded interpreter. The calculator, phonon and band-structure dialogs each carry an **Execution Environment** selector that routes the job to any Conda environment or interpreter path (Section 8.3). This is how you keep, say, a GPAW environment separate from the one Calango itself uses.

1.2.3 The environment file

At launch Calango reads an environment file — `~/.env` by default — and exports `MP_API_KEY`, the Materials Project API key, into the process environment. A key already present in your shell takes precedence at startup.

Point Calango at a different file, or re-read it on demand, in **Edit ⟩ Preferences**. The dialog reports whether the file exists and whether it actually defines a key. The same controls are mirrored in the **Materials Project** tab of the database browser.

1.3 First launch

Calango opens with an empty workspace. Three quick ways to get something on screen:

1. **File ⟩ Open ⟩ Structure** and pick any structure file (Section 4.1 lists the formats).
2. **Build ⟩ From Database** and load one of the bundled presets — diamond, bulk and monolayer MoS_2, graphene, benzene, naphthalene, coronene.
3. Pass a file on the command line: `calango structure.cif`. Each file argument opens in its own tab; a `.calproj` argument restores a whole saved session.

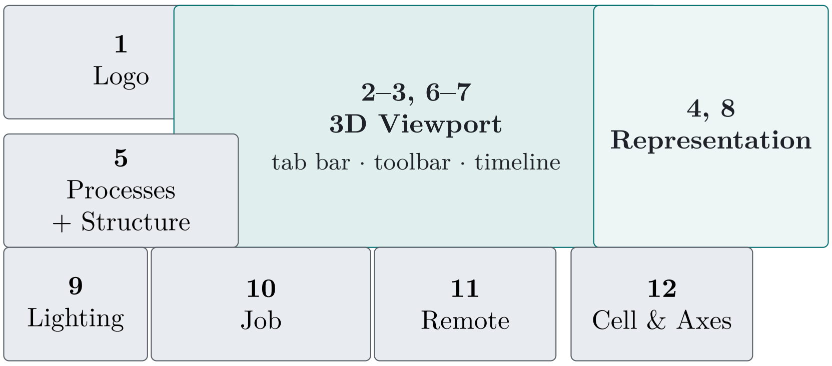
Once a structure is loaded, F frames it in the viewport and the **Structure** panel fills in with formula, cell parameters and symmetry.

CHAPTER 2

The Workspace

2.1 The 12-zone grid

Calango opens into a fixed default arrangement: a 4×3 grid of twelve zones around the central 3D viewport. Every zone is a Qt dock widget, so the layout is a starting point rather than a constraint — any panel can be resized, re-docked, tabbed with another, floated onto a second monitor, or hidden entirely.



Zone 1 — branding The Calango banner. Purely decorative; hide it from **View** if you want the vertical space.

Zone 5 — Processes and Structure The process manager (Section 2.5) sits above the structure information panel.

Zones 2–3, 6–7 — the viewport The document tab bar, the frame toolbar, the 3D canvas, and the playback timeline (visible only for trajectories). Covered in Chapter 3.

Zones 4 and 8 — Representation Appearance controls, spanning the full height of the right column (Chapter 7).

Zone 9 — Lighting Directional light editor.

Zone 10 — Job Console and live metric plots for the running calculation (Section 8.4).

Zone 11 — Remote Access SSH connection, cluster job submission and queue monitoring (Chapter 9).

Zone 12 — Unit Cell & Axes Cell wireframe and axes triad styling.

Tip
Your layout is saved when you quit and restored on the next launch, including floating panels and window geometry. Every panel has a toggle at the bottom of the **View** menu, so a panel you closed by accident is one menu click away.

2.2 Documents, tabs and the timeline

Each structure or trajectory you open occupies its own *tab* above the viewport, and each tab carries its own undo history. All tabs share one accelerated viewport — switching tabs changes which document the views observe, so display settings stay consistent and no OpenGL context is recreated.

When a document contains more than one frame, the **playback timeline** appears below the viewport:

Transport buttons first, previous, play/pause, next, last.

Scrubber a tick-marked slider over the frame range.

Rate playback speed in frames per second, 0.1–120, default 15.

Trajectories arrive from several places: opening a multi-frame file, finishing an MD or relaxation run, generating displaced structures in the phonon builder, or producing a noise ensemble. During a running simulation the timeline grows in real time as frames stream in (Section 8.5).

2.3 The Structure panel

The **Structure** panel reports what Calango knows about the active document:

Formula Hill-ordered chemical formula.

Atoms, Bonds Counts, with bonds as currently perceived.

a, b, c Cell vector lengths in Å, to two decimals.

α, β, γ Cell angles in degrees.

Cell volume, Periodic Volume and per-axis periodicity.

Tolerance The spglib symmetry tolerance (*symprec*) used for the three fields below, in Å, up to four decimals. Default 0.001. Raise it for structures with numerical noise — a relaxed cell often needs 0.01 or more before its true symmetry is recognised.

Space group, Point group, Crystal system Detected symmetry, e.g. `Fd-3m (227)`, `m-3m`, `cubic`.

Requires
Symmetry detection needs `spglib` and a defined unit cell.

2.4 Projects and session storage

A *project* is one `.calproj` file that restores an entire session: every tab with its structures and trajectory frames, the viewport colour mapping, and the job console with its recorded metric series.

File ⟩ Project Workspace ⟩ Open Project Ctrl+Shift+O. Replaces the current session, warning first if tabs are open.

File ⟩ Project Workspace ⟩ Save Project Ctrl+S.

File ⟩ New Workspace Ctrl+N. Closes everything and starts clean.

Calango tracks unsaved changes. Every undoable edit, tab close and job launch marks the session dirty; saving or loading a project clears the flag. Quitting with unsaved changes asks whether to **Save**, **Discard** or **Cancel** — cancelling (or cancelling the save dialog that follows) leaves the application open.

2.4.1 The managed temporary directory

Simulation jobs and generated ensembles need somewhere to put their working files. Once a project has been saved, Calango places them in a `.calango_tmp/` folder *next to the `.calproj` file*, one timestamped subdirectory per task:

```
my_study.calproj
.calango_tmp/
    job_20260721_143255/     run.py, structure.extxyz, md.traj, logs
    noise_20260721_144012/   perturbed.extxyz
    job_20260721_150130/     bands.json, pdos.json
```

Until a project has been saved, the same directories are created under the per-user application data location instead. Either way the process manager keeps a live link to each one.

2.5 The process manager

The compact **Processes** panel in zone 5 lists every background task of the session — local calculations, remote submissions, band-structure runs, noise ensembles — with its name, colour-coded status (*queued*, *running*, *completed*, *failed*) and start time.

Two buttons act on the selected task:

Open Folder Reveals the task's working directory in your file manager, for the raw logs and any files Calango does not import.

Load Result Brings the result back into the workspace. Double-clicking the row does the same. Calango inspects the directory and opens whatever it finds: band structure and PDOS data open in the electronic-structure viewer, trajectories and final geometries open as tabs.

The point of keeping these links is that a finished calculation stays one click from further analysis. A completed MD run can be re-loaded weeks later and fed to the RDF, distribution or dataset tools without recomputing anything.

CHAPTER 3

The 3D Viewport

The viewport is an OpenGL 3.3 core-profile canvas using instanced rendering — one draw call per mesh type — so structures with tens of thousands of atoms stay interactive.

3.1 Interaction modes

What a left-drag does depends on the active *interaction mode*. The six modes are the first group of buttons on the frame toolbar, each with a single-letter shortcut:

Key	Mode	Left mouse button
R	Rotation	Drag orbits the camera around the structure. The default.
T	Translation	Drag pans the scene.
S	Selection	Drag draws a rubber-band box and selects every atom inside it.
I	Insertion	Click empty space to add an atom; drag from one atom to another to bond them.
D	Distance	Click two atoms to measure their separation.
A	Angle	Click three atoms (vertex second) to measure the angle.

Three gestures work in *every* mode, so navigation never requires switching back:

Middle-drag, or Shift+left-drag Pan.

Wheel Zoom.

Double-click Reframe the structure.

The mouse cursor changes with the mode — an arrow for rotation, an open hand for panning, a crosshair for selection, insertion and measurement.

3.2 Selecting atoms

A left *click* (no drag) picks the atom under the cursor by ray-cast, in any mode. Ctrl+click (Cmd+click on macOS) toggles an atom in or out of the selection, so you can build up a set one atom at a time. Clicking empty space clears the selection.

In **Selection** mode a rubber-band drag selects every atom whose centre falls inside the box; holding Ctrl adds the boxed atoms to the existing selection instead of replacing it. With the viewport focused, Delete or Backspace removes the selected atoms and the bond network is rebuilt immediately.

The status bar reports the selection size, and several tools act on it: the bond-order buttons need exactly two atoms, the bond editor can pull a pair straight from the selection, and **Edit ⟩ Selection** can change the element of, or translate, whatever is selected.

3.3 Measuring distances and angles

Measurement modes accumulate clicks on atoms and draw the result directly on the canvas — markers on the chosen atoms, dashed connecting lines, and the value in a floating label. The same value is written to the status bar, for example:

```
Distance Si(0) - Si(4): 2.351 A
Angle O(2) - Si(0) - O(5): 109.47 deg
```

Distance Two atoms; result in Å to three decimals.

Angle Three atoms; the *second* atom clicked is the vertex. Result in degrees to two decimals.

Clicking empty space clears the current measurement, and clicking again after a completed one starts fresh. Dragging still orbits the camera, so you can rotate the structure between clicks to reach an atom hidden behind another. Measurements are cleared automatically when you switch modes or load a different structure.

3.4 Fixed-angle rotations

The toolbar's rotation cluster turns the scene by exact increments about the world axes — useful for reproducible figure orientations.

Angle step An editable spin box, 1–180°, default 15°.

X+ X– Y+ Y– Z+ Z– Counter-clockwise and clockwise rotation about each axis by exactly that step.

Each click animates smoothly over about 200 ms, and rapid clicks compose exactly: four presses of **Z+** at 90° return the scene to where it started. The axes triad follows the rotation, and picking, selection and measurements all stay consistent with what you see.

3.5 Framing, alignment and projection

F Frame the structure: centre it and back the camera off far enough to see all of it. Also on the toolbar and in **View ⟩ Alignment**.

Align with XY / XZ / YZ Look straight down the remaining Cartesian axis. Toolbar buttons, or **View ⟩ Alignment**. Alignment clears any accumulated fixed-angle rotation, so the result is a true axis-aligned view.

O Toggle orthographic versus perspective projection. The orthographic frustum is matched to the perspective field of view at the camera's target distance, so the structure keeps its apparent size across the switch — and zoom keeps working in both.

Tip
Orthographic projection is what you want for figures where lattice directions must read as parallel: perspective foreshortening makes a supercell look subtly sheared.

3.6 Inserting atoms and bonds

In **Insertion** mode (I) the toolbar's element button — which shows the active symbol, `C` by default — selects what gets placed. Click it to choose any element from the graphical periodic table.

Click on empty space Inserts one atom of the active element there. The atom lands on the plane through the camera's target point facing the viewer, so what you click is where it appears.

Drag from one atom to another Creates a bond between them, added as a manual bond override.

Click on an existing atom Selects it, as in any other mode.

Both operations are single undo steps.

Note
Because inserted atoms land on the camera-target plane, the depth of a new atom depends on the current view. Place atoms from an aligned, orthographic view when the exact coordinate matters, or insert roughly and then set exact coordinates with **Edit ⟩ Add Atom**, which takes numeric x, y, z.

3.7 Visual effects

View ⟩ Visual Effects opens a modeless dialog — the scene stays interactive while you tune it — with two depth cues.

3.7.1 Distance fog

Fades distant geometry into the background colour, which makes the depth ordering of a thick slab or a large nanoparticle immediately readable.

Mode Linear (start → end) fades uniformly between two distances; **Exponential (density)** follows $e^{-\rho d}$.

Start distance, End distance The linear ramp, in Å. Defaults 15 and 80.

Density The exponential coefficient, default 0.03.

The fog colour tracks the viewport background automatically, so geometry always fades into whatever backdrop you have chosen.

3.7.2 Depth of field

A post-processing pass that blurs geometry away from the focal plane, in proportion to its circle of confusion.

Blur strength Maximum blur radius in pixels, 1–20.

Focus range The depth band that stays sharp, in Å.

Focus offset Shifts the focal plane relative to the camera target, in Å.

> **Note**
> Depth of field renders the scene into an offscreen buffer before compositing, which means multisample anti-aliasing is unavailable while it is enabled. For final figures, decide which of the two matters more — crisp edges or the depth cue.

CHAPTER 4

Getting Data In and Out

4.1 Supported formats

All structure I/O goes through `ase.io`, so Calango reads and writes every format ASE knows. The file dialog offers the common ones directly:

Family	Extensions
XYZ	`.xyz`, `.extxyz`
Crystallography	`.cif`
VASP	`POSCAR`, `CONTCAR`, `.vasp`
ASE	`.traj`
Quantum ESPRESSO	`.in`, `.pwi`, `.pwo`, `.out`
CASTEP	`.cell`
LAMMPS	`.data`, `.dump`, `.lammpstrj`
Gaussian	`.gjf`, `.com`
SHELX	`.res`

Note

Extended-XYZ per-atom arrays are imported as *scalar fields* and *vector fields*. A `charges` or `forces` column becomes selectable in the **Custom property** colour mode, and `forces` or `momenta` enable the force and velocity arrow overlays. This is the simplest route for visualising per-atom data computed elsewhere: write it as an extxyz column and open the file.

4.2 Opening structures and trajectories

File 〉 Open 〉 Structure Ctrl+O.

File 〉 Open 〉 Trajectory Ctrl+T.

Both entries read *every* frame in the file. A single-frame file opens as a plain structure tab; a multi-frame file opens as a trajectory with the timeline active. In practice the two entries differ only in the file filter, so either works for either kind of file.

Opening a `.calproj` file — from the command line, your file manager, or the structure dialog — restores the saved session rather than loading a geometry.

4.3 Saving structures and trajectories

File ⟩ Save ⟩ Structure As Ctrl+Shift+S. Writes the currently displayed frame. The format follows the extension you type.

File ⟩ Save ⟩ Trajectory As Ctrl+Shift+T. Writes all frames of the active document as extended XYZ, multi-frame XYZ, ASE `.traj`, or multi-model PDB.

> **Tip**
> Extended XYZ round-trips more than plain XYZ — cell, periodicity and per-atom arrays all survive. When in doubt, save `.extxyz`.

4.4 The database and preset browser

Build ⟩ From Database opens a two-tab browser.

4.4.1 Presets

Seven benchmark structures ship with Calango, each annotated with a suggested calculator:

Preset	File	Suggested calculator
Diamond (bulk C)	`diamond.vasp`	MACE-MP-0
MoS_2 2H (bulk)	`mos2_2h_bulk.vasp`	MACE-MP-0
Graphene monolayer	`graphene.vasp`	MACE-MP-0
MoS_2 1H (monolayer)	`mos2_1h_monolayer.vasp`	MACE-MP-0
Benzene	`benzene.xyz`	MACE-OFF (or EMT)
Naphthalene	`naphthalene.xyz`	MACE-OFF
Coronene	`coronene.xyz`	MACE-OFF

Select a row and press **Load Selected**, or double-click it.

4.4.2 Materials Project

The second tab fetches structures by Materials Project ID.

API key Masked field, pre-filled from your saved setting or from `MP_API_KEY` in the environment. It is stored as you type.

Material ID For example `mp-149` (silicon). Pressing Enter triggers the fetch.

.env file Shows the current environment file path, with **Browse...** to point elsewhere and **Reload** to re-import the key (Section 1.2.3).

Fetch Structure Downloads the entry and opens it in a new tab, reporting the formula and atom count.

> **Requires**
> Needs a network connection and a personal API key from `https://materialsproject.org/api`.

CHAPTER 5

Building Structures

Everything in the **Build** menu produces a new structure, either replacing the active one or opening in a new tab. All of it runs through ASE, so a working Python environment is a prerequisite throughout this chapter.

5.1 Supercells

Edit ⟩ Unit Cell ⟩ Create Supercell repeats the active structure along its lattice vectors. Three spin boxes, **Repeat along a/b/c**, each 1–20, default $2 \times 1 \times 1$. The operation is undoable and requires a defined cell along every repeated direction.

5.2 The surface slab wizard

Build ⟩ Surface Slab cleaves a slab from the active bulk structure in three guided stages. Unlike a plain Miller-index dialog, each stage shows you the consequence of your choice before you commit.

5.2.1 Stage 1 — surface orientation

Set the Miller indices $(h\,k\,l)$ with three spin boxes (-9 to 9), or work the other way round: drag the red **u** and green **v** handles on the interactive lattice canvas to any two lattice points, and Calango computes the nearest integer $(h\,k\,l)$ for the plane they span. The header shows the current vectors and resulting indices live, and the info line reports $|u|$, $|v|$ and the angle between them.

Rotate the axonometric view by dragging; only non-collinear handle pairs are accepted. **Next** unlocks once a test cut succeeds — $(0\,0\,0)$ is rejected as not a plane.

5.2.2 Stage 2 — cut, thickness and terminations

Calango builds a reference stack of eight bulk repeats and clusters the atoms into atomic layers (0.10 Å tolerance). The cross-section canvas draws each layer as a dashed line labelled with its z coordinate, with the current selection band highlighted between an orange *top termination* and a teal *bottom termination*.

Click a layer Moves whichever boundary is nearer the click, so you can set either termination directly.

Layers in slab Sets the count numerically.

Thickness The same choice in Å; it snaps to whole atomic layers above the bottom termination.

The info line reads back the selection, for example *layers 1–8 of 16 (8 layers, 12.45 Å)*.

5.2.3 Stage 3 — vacuum and preview

Top vacuum and **Bottom vacuum** (0–80 Å, default 10 each) pad the cell along the surface normal. **Symmetric vacuum (centered slab)**, checked by default, mirrors the top value to the bottom. A live 3D preview shows the finished slab, with atom count, slab thickness and total cell height. **Finish** inserts it as a new tab labelled, for example, *(1 1 1) slab, 8 layers*.

5.3 Nanomaterials

Build ⟩ Nanomaterial Builder generates four families of low-dimensional carbon and TMD structures. Pick the type at the top; the form below changes to match.

Graphene sheet Lattice constant a (default 2.46 Å), repeats $n_x \times n_y$ (default 4×4), vacuum (default 10 Å).

Graphene nanoribbon Width and length in unit cells (default 6 each), **Edge type Zigzag** or **Armchair**, and **Hydrogen-terminate edges** (on by default).

Carbon nanotube Chiral indices n and m (default 6, 6), length in unit cells, C–C bond length (default 1.42 Å), vacuum (default 8 Å). $(0, 0)$ is rejected.

TMD monolayer (MX_2) An editable formula combo — `MoS2`, `MoSe2`, `WS2`, `WSe2`, `MoTe2`, `NbSe2`, or anything else you type — **Phase** `2H` or `1T`, lattice constant (default 3.16 Å), X–X thickness (default 3.19 Å), repeats and vacuum.

5.4 Metallic nanoparticles

Build ⟩ Metallic Nanoparticle (Wulff) builds clusters two ways. Choose the element from the periodic table, then a mode.

5.4.1 Wulff construction

The equilibrium shape of a crystal is the one minimising total surface energy at fixed volume; the Wulff construction produces it from the relative surface energies of the exposed facets.

Facet table One row per facet: h, k, l and γ (relative surface energy). Defaults are $(111) = 1.0$, $(100) = 1.1$, $(110) = 1.2$. **Add Facet** and **Remove Selected** edit the list.

Target atoms The requested size, default 200. The construction lands on the nearest achievable closed shape.

Size rounding `closest`, `above` or `below`.

Lattice `fcc`, `bcc` or `sc`.

Only the *ratios* of the γ values matter. Lowering $\gamma(111)$ relative to $\gamma(100)$ grows the (111) facets and drives an fcc metal from a cube towards a truncated octahedron.

5.4.2 Spherical cluster

Carves every atom within a cutoff radius of the centre out of a bulk lattice — `fcc`, `bcc`, `sc` or `hcp`. Set **Radius** (default 10 Å). Useful when you want a specific diameter rather than a thermodynamic shape.

Both modes centre the cluster with 6 Å of vacuum and clear periodicity. **Lattice constant** may be left at **ASE default** to use the tabulated value for the element.

5.5 Special quasirandom structures

Build ⟩ Special Quasirandom Structure (SQS) builds a substitutional alloy whose short-range order approximates a truly random solid solution — the standard way to model a disordered alloy in a small periodic cell.

Supercell Repetitions $n_1 \times n_2 \times n_3$, default $2 \times 2 \times 2$.

Replace element Whose sites form the alloy sublattice.

Composition Target occupancies as symbol:fraction pairs, e.g. `Cu:0.75, Au:0.25`. Fractions are normalised and rounded to whole atoms by largest remainder.

First shell cutoff, Second shell cutoff The pair shells whose Warren-Cowley parameters are driven toward zero. Defaults 3.2 and 4.8 Å; set the second to **off** to use one shell.

MC steps, Random seed Anneal length and reproducibility.

If `icet` is importable, Calango uses its `generate_sqs`; otherwise it falls back to a built-in simulated annealer that minimises $\sum \alpha^2$ over the chosen shells using only NumPy and ASE neighbour lists. The resulting tab is labelled with the backend that produced it, and the internal backend also reports its residual objective — a value near zero means the decoration is essentially ideal.

> **Tip**
> Verify the result with **Analysis ⟩ Warren-Cowley analysis** (Section 11.6): a good SQS shows $\alpha \approx 0$ for the shells you optimised.

5.6 Normal modes and phonons

Build ⟩ Normal Modes / Phonon Builder sets up a finite-displacement vibrational calculation. Calango detects whether the structure is periodic and adapts:

Periodic Supercell finite displacements via `ase.phonons`, giving dispersion and DOS.

Isolated molecule Γ-point normal modes via `ase.vibrations`, giving frequencies and per-mode animation trajectories that Calango opens directly.

5.6.1 Parameters

Mode **Run vibrational calculation**, or **Generate displaced structures only (new tab)** to export the displacement set for an external DFT code.

Supercell (a × b × c) Default $2 \times 2 \times 2$, periodic structures only.

Displacement δ Default 0.01 Å.

Displaced structures A read-only count: $6N + 1$ for N supercell atoms.

Calculator EMT, Lennard-Jones or MACE — deliberately limited to calculators needing no external binary or pseudopotentials.

Band-path points, DOS k-grid Sampling for the dispersion and the phonon density of states (defaults 100 and 20).

Like the calculator dialog, the generated ASE script is shown in an editable pane and can be saved with **Save Script. . .** .

Outputs are Γ-point frequencies in the job console, plus `phonon_bands.csv` and `phonon_dos.csv` in the job directory.

> **Note**
> Displacements are applied to every atom of the expanded supercell ($6N_{\text{supercell}} + 1$ configurations), following the standard finite-displacement recipe without symmetry reduction. For large supercells this is the dominant cost; symmetry-reduced displacement sets are a planned improvement.

5.7 Structure perturbation and noise

Build ⟩ Structure Perturbation / Noise adds random displacements — for shaking a structure out of a symmetric saddle point, or for generating training ensembles for machine-learning potentials.

Distribution **Gaussian (normal)** or **Uniform**.

Amplitude Default 0.05 Å. For Gaussian this is σ per component; for uniform it is the half-width.

Random seed Default 42, for reproducibility.

Perturb atomic positions On by default.

Perturb unit cell vectors (random strain) Atoms follow the cell affinely, preserving fractional coordinates.

Frames 1 perturbs in place; more generates a trajectory.

Accumulation **Independent** draws each frame from the original structure; **Cumulative** performs a random walk from the previous frame.

A multi-frame run opens as a trajectory whose frame 0 is the unperturbed original, is registered in the process manager, and is checkpointed to `perturbed.extxyz` in the managed temporary directory (Section 2.4.1) — so the ensemble can be reloaded later and fed to the analysis or dataset tools.

CHAPTER 6

Editing Structures

Every operation in this chapter is a single undo step. **Edit ⟩ Undo** (Ctrl+Z) and **Edit ⟩ Redo** (Ctrl+Shift+Z) walk a per-document history up to 50 states deep.

6.1 Atoms

Edit ⟩ Add Atom Ctrl+Shift+A. Choose the element from the graphical periodic table and type exact x, y, z coordinates. For placing atoms by eye instead, use Insertion mode in the viewport (I).

Edit ⟩ Selection ⟩ Change Element of Selection Replaces the species of every selected atom, again via the periodic table.

Edit ⟩ Selection ⟩ Translate Selection Shifts the selection by a Δx, Δy, Δz vector in Å.

Edit ⟩ Delete Selected Atoms Delete. Removes the selection; with the viewport focused in Selection mode, Backspace does the same.

Deleting atoms re-indexes everything that referenced them — bond overrides, bond orders and per-atom fields all follow correctly.

6.2 The periodic table selector

Wherever an element must be chosen, Calango opens a graphical periodic table: $Z = 1$ to 118 in the standard 18-group layout, f-block below, coloured by chemical family. One click selects and closes.

6.3 Bond perception

By default Calango perceives bonds by distance: atoms i and j are bonded when

$$d_{ij} < t \cdot (r_{\text{cov}}(i) + r_{\text{cov}}(j)),$$

with covalent radii from Cordero *et al.* and a tolerance t you control. With a periodic cell the minimum-image convention applies, so bonds across cell boundaries are found and drawn correctly.

Edit ⟩ Bond Editor (Ctrl+B) exposes both the automatic rule and manual overrides:

Automatic bond detection Turn it off to draw *only* manually added bonds — the right choice for coarse-grained or schematic figures.

Covalent cutoff multiplier The tolerance t, 0.50–2.50, default 1.15. Applied live, so you can dial it until the picture is right.

Manual Bonds Two 1-based atom-index spin boxes, or **From Selection** to fill them from a two-atom viewport selection. The info line shows the current pair's distance next to the automatic cutoff for that pair, which makes it obvious whether a bond is borderline.

Add Bond Forces a bond regardless of distance.

Suppress Bond Removes an automatically perceived bond.

Active overrides Lists every override as *added* or *suppressed*, with the pair and its distance. **Clear Override** and **Clear All** undo them.

Overrides live on the structure and are saved in the project file.

6.4 Bond orders

Calango never guesses multiple bonds. Every perceived bond starts as a single bond, and orders are assigned deliberately — distance-based perception of double and triple bonds is unreliable, particularly for inorganic and metallic systems where short contacts are common.

To assign one, select exactly two atoms in the viewport (Ctrl+click the second) and use the **Bond order** buttons in the **Representation** panel: **Single**, **Double**, **Triple**. The buttons are disabled until a pair is selected.

An order of two or three renders as that many parallel cylinders, and also forces the bond to exist even if the pair lies outside the automatic cutoff. Orders persist in project files.

> **Note**
> A consequence worth knowing: molecules like benzene or CO_2 render with all-single bonds when first opened. Assign the orders you want to display; they are a presentation property, not an input to any calculation.

CHAPTER 7

Appearance and Representation

7.1 The Representation panel

7.1.1 Representation mode

Mode selects how atoms and bonds are drawn:

Ball-and-Stick Spheres at a fraction of the covalent radius, bonds as cylinders. The default.

Space-filling (CPK) Spheres at approximately the van der Waals radius; bonds are hidden inside the atoms.

Wireframe Bonds only, drawn as lines with points at the atom positions — the cheapest mode for very large systems.

Atom radius and **Bond width** scale everything globally, 0.20–3.00, each with a slider and a synchronised spin box for exact values.

7.1.2 Colouring atoms

Color by offers four mappings, applied consistently to atoms and to their halves of each bond:

Element (CPK) The Jmol palette, with per-element overrides.

Coordination number (CN) Discrete coordination numbers on a gradient.

Generalized CN (GCN) Continuous generalised coordination numbers — excellent for distinguishing terraces, steps, edges and vertices on nanoparticles and slabs.

Custom property Any per-atom scalar field the structure carries: charges, force magnitudes, extended-XYZ columns, or a computed field such as `local_entropy`. **Property** selects which.

For the three scalar modes, **Gradient** chooses the colour map — **Viridis**, **Plasma**, **Turbo**, **Inferno**, **Magma**, **Cividis**, **Hot**, **Afmhot**, **Coolwarm**, **Rainbow** — and **Invert palette** reverses the scalar-to-colour mapping, matching matplotlib's `_r` variants. **Range** reads back the min and max of the active field, so the legend is never ambiguous.

> **Tip**
> Viridis and Cividis are perceptually uniform and colour-vision safe; Coolwarm is the right choice for signed quantities such as charges or potentials, where the zero point should read as neutral. Rainbow is included because reviewers sometimes ask for it, not because it is a good default.

Colour mapping is recomputed for every frame during trajectory playback, so CN, GCN and property maps stay live throughout an animation.

7.1.3 Bonds and vectors

Gradient bond coloring Blends each bond smoothly from one atom's colour to the other's, instead of the classic half-and-half split.

Force vectors, Velocity vectors Draw per-atom vectors as lit 3D arrows. Each is enabled only when the structure actually carries that field; the tooltip explains what is missing when it does not.

Vector scale Arrow length in Å per field unit, 0.05–200, default 10.

7.1.4 Per-element styling

Element Settings... opens a table of the elements present in the structure. Each row offers a colour swatch and a radius scale (10–300 %, where exactly 100 % means "no override"), plus a **Reset** button. **Reset All** clears every override at once.

Background sets the viewport background colour. Distance fog, if enabled, automatically fades into whatever you choose.

7.2 Unit cell and axes

The **Unit Cell & Axes** panel (zone 12):

Show unit cell Also on **View** 〉 **Overlays**.

Cell color Wireframe colour.

Cell line width 1.0–8.0, default 2.0. A width of 1 draws plain GL lines; anything larger renders the edges as lit tubes.

Show axes triad The corner orientation indicator.

Axes style **Cartesian (X, Y, Z)** or **Lattice vectors (a1, a2, a3)**.

Axes size On-screen size of the triad, 48–240 px.

> **Note**
> Core-profile OpenGL clamps hardware line width to one pixel, which is why thick cell edges are drawn as tube geometry rather than wide lines. The tubes are lit like the rest of the scene, so they also read correctly in ray-traced exports.

7.3 Lighting

The **Lighting** panel (zone 9) edits up to four directional lights with Blinn-Phong shading. The default studio setup is three lights: a warm key carrying the speculars, a cooler fill from the opposite side, and a back (rim) light that separates atom silhouettes from the background.

Select a light in the list, then edit:

Direction (view space) Three components, -5 to 5. Because directions are in view space, the lighting stays fixed relative to the viewer as you orbit — the structure turns, the studio does not.

Ambient, Diffuse, Specular Per-light colours.

Add appends a fill light from the opposite side; **Remove** deletes the selected one. At least one light always remains.

CHAPTER 8

Running Simulations

8.1 The calculation dialog

Simulation ⟩ New Calculation (ASE) (Ctrl+R) is the main entry point. The left half is a form; the right half is the ASE script that form produces, live and editable.

8.1.1 Calculators

Calculator	Notes
EMT	Effective medium theory. Fast, ships with ASE, useful for testing a workflow before committing compute.
Lennard-Jones	Ships with ASE.
Quantum ESPRESSO	DFT; needs `pw.x` and pseudopotentials.
VASP	DFT; needs a license and POTCARs.
MACE	Machine-learning potential; needs `mace-torch`.
GPAW	DFT as a Python package.
SIESTA	DFT; needs the `siesta` binary and pseudopotentials.
ORCA	Quantum chemistry; needs the ORCA binaries.

The DFT entries generate working script skeletons with clearly marked `EDIT ME` lines where you must supply pseudopotential paths or launch commands.

8.1.2 Tasks

Single-point energy One energy and force evaluation.

Geometry optimization (BFGS) Controlled by **Force convergence (fmax)** (default 0.05 eV/Å) and **Max optimization steps** (default 200). Writes `opt.traj`.

Molecular dynamics See below. Writes `md.traj`.

8.1.3 Molecular dynamics ensembles

Ensemble	Thermostat / barostat	Key parameters
NVE	Velocity Verlet	timestep
NVT	Langevin (default)	temperature, friction
NVT	Andersen	temperature, collision probability
NVT	Berendsen	temperature, τ_T
NVT	Nosé–Hoover chain	temperature, τ_T
NPT	Berendsen	temperature, pressure, τ_T, τ_P
NPT	Nosé–Hoover / Parrinello–Rahman	temperature, pressure, τ_T, τ_P

Shared parameters: **Temperature** (default 300 K), **Timestep** (default 1 fs), **MD steps** (default 1000), **Thermostat coupling** (default 100 fs), **Barostat coupling** (default 1000 fs) and **Pressure** (default 0 GPa). Fields enable themselves only for the ensembles that use them.

> **Note**
> Every MD run initialises velocities from a Maxwell-Boltzmann distribution and then removes the net centre-of-mass momentum, so the system does not drift as a whole. For isolated (non-periodic) systems the net angular momentum is removed as well. Without this, a slow overall translation contaminates every subsequent analysis — diffusion coefficients most of all.

8.1.4 DFT and ORCA parameters

DFT calculators expose **Plane-wave cutoff** (default 550 eV) and a **k-point grid** (default $4 \times 4 \times 4$).

ORCA has its own group:

ORCA method Editable: `B3LYP`, `PBE0`, `r2SCAN`, `wB97X-D3`, `HF`, `MP2`, or any ORCA keyword you type.

ORCA basis Editable: `def2-SVP`, `def2-TZVP`, `def2-TZVPP`, `cc-pVDZ`, `cc-pVTZ`, ...

Charge -10 to 10.

Multiplicity $2S + 1$, 1 to 11.

Solvation **None (gas phase)**, **CPCM** or **SMD**, with **Solvent** naming the medium (water, acetonitrile, toluene, ...).

ASE writes the ORCA `.inp` file into the job directory and parses the output. The generated script contains an `EDIT ME` line for the path to your `orca` binary.

8.1.5 MACE models

MACE model **MACE-MP-0 (universal, materials)**, **MACE-OFF (universal, organic molecules)**, or **Custom trained model (file)**.

MACE model size `small`, `medium` (default), `large`.

Custom model file A `.model` or `.pt` checkpoint.

MACE device `cpu`, `cuda` or `mps`.

Foundation models download automatically on first use and are cached in `~/.cache/mace`.

8.2 The script editor

The right-hand pane always shows the complete, standalone ASE script the form describes — syntax-highlighted and fully editable.

Start typing in it and Calango shows *edited — form sync paused* in amber: from then on the form no longer overwrites your text, so hand edits are never silently lost. **Regenerate** discards them and re-syncs. **Save Script...** writes the current text to a `.py` file.

Tip
This is the intended escape hatch for anything the GUI does not expose. Configure what you can in the form, then edit the script for the rest — constraints, custom observers, a different optimiser. The script is ordinary ASE and runs unmodified on a cluster.

8.3 Execution environments

The **Execution Environment** group decides which Python actually runs the job — independent of the interpreter Calango embeds.

Leave it empty The job uses the embedded interpreter. The status line names it.

Env Folder... Point at a Conda environment directory; Calango finds `bin/python` inside it.

Python... Point directly at an interpreter.

The status line confirms the resolution live, turning red when no interpreter can be found at the given path. The setting is remembered between sessions, and the same selector appears in the phonon builder and the band-structure dialog.

Tip
This is how you keep heavyweight solver stacks isolated: a GPAW environment for band structures, a `mace-torch` environment with CUDA for ML potentials, and a minimal ASE environment for Calango itself.

8.4 The Job panel

Jobs run as separate processes in their own directory, so a crashing solver cannot take the GUI with it. The **Job** dock (zone 10) has five tabs.

8.4.1 Log

A status label (*Idle*, *Running...*, *Finished (exit N)*, *Crashed*), a progress bar, a **Kill** button, and the live output. Standard error is coloured red, the start line blue, and a clean exit green.

8.4.2 Metric plots

Four tabs plot job observables live, parsed from markers the generated scripts print:

Tab	Quantity	Unit	Reference line
Energy	Total energy	eV	—
Temperature	Ionic temperature	K	thermostat setpoint
Force	Max $\lvert F \rvert$	eV/Å	—
Pressure	Scalar pressure	GPa	barostat setpoint

Each plot shows the running series with the latest value read out numerically, and a dashed reference line at the setpoint for thermostatted or barostatted runs — so drift away from the target is visible at a glance. Every tab has its own **Export Data...** button writing CSV or whitespace-separated `.dat`. The series are saved in the project file, so a reopened project shows the plots from the run that produced it.

8.5 Live trajectory streaming

MD runs and geometry relaxations do not make you wait. The moment the job starts, Calango opens a trajectory tab marked *(live)* and streams each newly computed geometry into it as the simulation produces it.

- The timeline grows as frames arrive, and follows the newest frame automatically.
- Scrub back to an earlier frame and the view stays there — Calango stops following so you can inspect while the run continues.
- Everything else works normally on the partial trajectory: measurement, colour mapping, representation changes.
- When the job finishes the *(live)* marker is dropped and the tab becomes an ordinary trajectory.

Relaxations stream every step; MD streams at an interval chosen so a run produces at most about 400 streamed frames, keeping long simulations responsive. The complete trajectory is still written to `opt.traj` or `md.traj` in the job directory.

8.6 What happens when a job finishes

Calango inspects the job directory and does the useful thing:

- A live-streamed run keeps the tab it already built.
- An electronic-structure run opens the band/PDOS viewer (Chapter 10).
- A trajectory (`md.traj`, `opt.traj`) opens in a new tab.
- Otherwise, if a final geometry exists (`optimized.extxyz`), Calango offers to load it.

The process manager entry switches to *completed* or *failed*, and its directory link remains available for the rest of the session and beyond.

8.7 The MLIP dataset manager

Simulation ⟩ Dataset Manager (MLIP) assembles training datasets for machine-learning interatomic potentials from the trajectories you have generated — MD runs, noise ensembles, relaxation paths — and partitions them reproducibly.

8.7.1 Loading structures

Add Files... accepts multiple files of mixed provenance: `.extxyz`, `.traj`, `.cif`, VASP files, ASE databases. Each is listed with its frame count and the chemical formulas it contains, so a heterogeneous dataset stays legible. The summary line reports the total.

8.7.2 Splitting

Training, Validation Percentages; **Test** is the remainder, shown live.

Random seed Default 42.

The split is deterministic: the same frame count, percentages and seed always produce exactly the same partition, and the three subsets are disjoint and cover every frame.

8.7.3 Query-by-Committee ensembles

Active learning needs an ensemble of models trained on different data, so their disagreement can be used as an uncertainty estimate.

Committee members 1 exports a single dataset; $N > 1$ adds `committee_01`...`committee_N` subdirectories, each with its own training set.

Diversity How the members are made to differ:

- **Independent splits (seed + k)** re-splits the non-test pool with a different seed per member, keeping one common test set.
- **Bootstrap resampling of the training set** draws each member's training set with replacement — the classical bagging construction, which reuses roughly 63 % of the original frames per member.

8.7.4 Export

Extended XYZ (MACE-ready) Writes `train.extxyz`, `valid.extxyz` and `test.extxyz` (plus committee subdirectories), the layout MACE and most training scripts expect.

ASE database (.db) One SQLite database per subset.

> **Note**
> Export goes through `ase.io` directly rather than Calango's internal structure model, specifically so that energies, forces and stresses attached to the frames survive into the training files. A dataset exported from a finished MD run carries the labels a potential needs to train on.

CHAPTER 9

Remote HPC Execution

The **Remote Access** panel (zone 11) submits calculations to a cluster over SSH, monitors the queue, and brings the results back — without leaving Calango.

> **Requires**
> Needs `paramiko` in Calango's Python environment, and SSH access to a cluster running SLURM, PBS or SGE.

> **Note**
> All network traffic runs in a separate helper process driven over a JSON protocol, mirroring how local jobs are isolated: the interface never blocks on the network, and a hung connection can always be resolved by killing the helper rather than the application.

9.1 Connection

The **Connection** tab holds the cluster credentials:

Host, Port Hostname and port (default 22).

User Your account name on the cluster.

Auth **SSH key** or **Password**.

Key file Path to a private key, e.g. `~/.ssh/id_rsa`. Leave empty to let your SSH agent or default keys authenticate.

Password The password, or the passphrase for an encrypted key.

Remote dir Working directory on the cluster, relative to `$HOME` (default `calango_jobs`).

Test Connection Verifies everything, reports your remote `$HOME`, and auto-detects the scheduler.

> **Requires**
> Passwords and key passphrases are kept in memory only — never written to disk, never placed on a command line where `ps` could see them. Everything else (host, user, key path, scheduler settings) is remembered between sessions.

9.2 Scheduler settings

The **Scheduler** tab describes the batch job:

Scheduler **SLURM**, **PBS** or **SGE**.

Queue Partition or queue name; empty uses the cluster default.

Tasks / cores Requested cores.

Walltime `HH:MM:SS`, default `01:00:00`.

Setup Shell lines executed before the payload — module loads, `conda activate`, anything your site needs:

```
module load python
source ~/venvs/ase/bin/activate
```

Submit Calculation... Opens the usual calculator dialog, then submits the result.

The generated wrapper carries the right directives for your scheduler — `#SBATCH`, `#PBS` or `#$` — and always redirects output to fixed file names so the monitor knows what to tail.

9.3 Submitting and monitoring

Simulation ⟩ **New Remote Calculation** (Ctrl+Shift+R) — or **Submit Calculation...** in the panel — runs the full sequence:

1. Stage `run.py` and `structure.extxyz` into a local job directory, exactly as a local run would.
2. Generate `job.sh` from the scheduler settings.
3. Upload the directory over SFTP, creating the remote path as needed.
4. Submit with `sbatch` or `qsub` and capture the job ID.
5. Start monitoring.

The **Queue & Logs** tab then shows the job ID and remote directory, the live queue state polled with `squeue` or `qstat`, and the remote standard output and error streamed incrementally into the console (errors in red). **Cancel Job** issues `scancel` or `qdel`.

9.4 Retrieving results

When the job leaves the queue, Calango downloads the results automatically — structures, trajectories, logs and CSV metric files — into the same local job directory the inputs came from. Trajectories and final geometries open directly in a new tab.

Download Results repeats the transfer manually, which is useful for pulling intermediate output from a long-running job without waiting for it to finish.

The task appears in the process manager like any other, so its directory stays one click away for later analysis.

CHAPTER 10

Electronic Structure

Simulation ⟩ Electronic Bands / PDOS computes a band structure along a high-symmetry k-path and, where the backend supports it, an element- and orbital-resolved projected density of states.

10.1 Setting up the calculation

Backend **Free electrons (ASE — bands only)**, **GPAW (DFT, bands + PDOS)** or **Quantum ESPRESSO (needs pw.x + pseudos)**. The GPAW entry is disabled when the package is not importable in the selected environment.

k-path Pre-filled with ASE's suggestion for the structure's Bravais lattice, e.g. `GXWKGLUWLK,UX`. Edit it freely; leave it empty to accept the suggestion.

k-points Total points along the whole path, default 80.

Valence electrons Electrons per cell, free-electron backend only.

PW cutoff Plane-wave cutoff, default 340 eV.

SCF k-grid Monkhorst-Pack n^3 grid for the self-consistent step, default 4.

Compute element/orbital PDOS (GPAW backend) Adds the projected DOS.

Environment The Conda environment or interpreter that runs the job (Section 8.3).

> **Tip**
> The free-electron backend needs nothing beyond ASE and finishes in seconds. It computes empty-lattice bands — the exact free-electron dispersion folded into your Brillouin zone — which is genuinely useful for checking that a k-path is the one you meant, and for teaching band folding, before committing a DFT run.

> **Requires**
> Real bands need a real solver: `gpaw` installed in the selected environment, or `pw.x` plus pseudopotentials for the Quantum ESPRESSO route. The QE script is a scaffold with an `EDIT ME` line for your pseudopotential set.

The job runs like any other — console, progress markers, process-manager entry — and writes `bands.json` and, when computed, `pdos.json` into the job directory. The viewer opens automatically on completion, and can be reopened any time with **Load Result** in the process manager.

10.2 Reading the plots

The viewer shows the band structure and, when present, the PDOS side by side on a shared energy axis.

10.2.1 Band structure

Energy versus k-path distance, with vertical lines and labels at every high-symmetry point (ASE's `G` is rendered as Γ). Spin channels are drawn in different colours. A dashed horizontal line marks the reference energy at $E - E_{\mathrm{ref}} = 0$.

10.2.2 Projected density of states

Density of states versus the same energy axis, one curve per element and orbital projection — `Si s`, `Si p`, `O p` and so on — accumulated over all atoms of each species and colour-keyed to the legend.

10.3 Controls and export

Fermi level The reference energy, pre-filled from the calculation. Plots show $E - E_{\mathrm{ref}}$, so shifting it moves the zero line — useful for aligning against a reference calculation or a measured spectrum.

E min, E max The energy window, defaults -10 and $+10$ eV.

Projections A checklist of the PDOS curves. Uncheck the ones that clutter the picture; the vertical scale re-normalises to what remains visible.

Export Bands... CSV or `.dat`: one row per k-point, with the path distance followed by every band (and spin channel).

Export PDOS... CSV or `.dat`: one row per energy, one column per projection.

> **Note**
> Exported bands use the same k-path distance as the x axis, so the columns drop straight into an external plotting script and reproduce the figure you see.

CHAPTER 11

The Analysis Toolbox

Every tool in this chapter opens from the **Analysis** menu. Most compute as soon as they open, run on a worker thread so the interface stays responsive, and export their data as CSV or whitespace-separated `.dat` — the format is chosen by the extension you type.

11.1 Radial distribution function

Analysis ⟩ Radial Distribution Function computes $g(r)$, the probability of finding an atom at distance r relative to a uniform density.

Element pair Two combos, each **Any** or a specific element. **Any**–**Any** gives the total RDF; a specific pair gives the partial.

r max Default 10 Å.

Bins Default 200.

Periodic boundary conditions Enabled and checked when the structure has a cell.

Frames For trajectories: **start**, **end** and **stride**, giving a frame-averaged $g(r)$ over any sub-range.

> **Note**
> With periodicity enabled, all periodic images within $r_{\max}$ are enumerated explicitly rather than relying on the minimum-image convention. The result is therefore valid beyond $L/2$ and correct for triclinic cells — both places where a naive implementation quietly produces artefacts.

11.2 Bond length and angle distributions

Analysis ⟩ Bond Length / Angle Distributions histograms either pair distances or three-body angles j–i–k within a cutoff.

Distribution **Bond length distribution** or **Bond angle distribution (j–i–k)**.

Species (pair / center–neighbor) Two element filters. For angles the first is the central atom.

Cutoff Default 3 Å.

Bins Default 90.

Angles are in degrees, lengths in Å; both handle periodic images exactly.

11.3 Coordination numbers (CN / GCN)

Analysis ⟩ Coordination Numbers (CN / GCN) computes per-atom coordination and generalised coordination numbers.

Neighbor cutoff **Covalent radii** × **tolerance** (default tolerance 1.15) or **Fixed cutoff radius** (default 3 Å).

Bulk CN reference The $\mathrm{cn}_{\mathrm{max}}$ normalising the GCN — 12 for fcc/hcp, 8 for bcc, 4 for diamond, or **Auto (max CN found)**.

Results appear as a per-atom table (index, element, CN, GCN) with a summary line giving the ranges and means. Two buttons push the result onto the structure: **Color Viewport by CN** and **Color Viewport by GCN**.

> **Note**
> The generalised coordination number weights each neighbour by its own coordination, which distinguishes sites that plain CN cannot: a terrace atom, a step-edge atom and a vertex atom on a nanoparticle can share the same CN but have clearly different GCN. This is what makes GCN such a good descriptor for catalytic activity.
> Periodic images are enumerated explicitly, so a primitive fcc cell correctly reports CN = 12 even though all twelve neighbours are images of the single atom in the cell.

11.4 Static structure factor

Analysis ⟩ Structure Factor S(q) computes $S(q)$ from the (frame-averaged) pair distribution via a Lorch-windowed Fourier transform.

q range Defaults 0.3 to 12 Å^{-1}.

q points Default 400.

g(r) r max Upper bound of the Fourier integral, default 10 Å. Larger values improve low-q fidelity.

g(r) bins Default 400.

Frames Start/end/stride, as for the RDF.

11.5 X-ray diffraction

Analysis ⟩ X-Ray Diffraction (XRD) simulates a powder pattern with the Debye scattering equation.

Wavelength Presets for Cu Kα (1.54056 Å), Co Kα, Mo Kα, Cr Kα, or **Custom** to type your own.

2θ range Defaults 10–90°.

Points Default 800.

Supercell repeat Periodic structures are repeated before the Debye sum, default 3. More repeats sharpen the Bragg peaks, but the cost grows as N^2 in atoms.

Unlike the other analysis dialogs, this one waits for you to press **Simulate**. The status line then reports which form factors were used: Waasmaier-Kirfel factors when every species is tabulated, otherwise a constant $f = Z$ approximation — in which case peak *positions* remain exact but high-angle *intensities* are approximate.

Two exports: **Export Curve...** writes the 2θ–intensity pattern, and **Export Peaks...** writes detected peaks (those above 2 % of the maximum) with their d-spacings from $d = \lambda/(2\sin\theta)$.

11.6 Warren-Cowley chemical order

Analysis ⟩ Warren-Cowley analysis quantifies chemical short-range order in multicomponent alloys through

$$\alpha_{ij} = 1 - \frac{p_{ij}}{c_j},$$

where p_{ij} is the probability that a neighbour of an i-type atom is of type j, and c_j is the overall concentration of j.

$\alpha = 0$ The ideal random alloy.

$\alpha < 0$ Ordering — unlike pairs preferred.

$\alpha > 0$ Clustering or segregation of like species.

Set **First shell cutoff** (default 3.2 Å) and, optionally, **Second shell cutoff** (default 4.8 Å). The dialog reports the concentrations, the mean coordination of each shell, and the full α_{ij} matrix for every ordered species pair, exportable as CSV.

> **Tip**
> A quick sanity check: a perfectly ordered B2 (CsCl-type) binary gives $\alpha_{AB} = -1$ in the first shell and +1 in the second, while a random decoration of the same lattice gives values near zero.

11.7 Local entropy

Analysis ⟩ Local Entropy Analysis computes the per-atom pair-entropy fingerprint of Piaggi and Parrinello, in units of k_B:

$$s_S^i = -2\pi\rho \int_0^{r_c} \left[\, g_i(r) \ln g_i(r) - g_i(r) + 1 \,\right] r^2 \, \mathrm{d}r,$$

with $g_i(r)$ the Gaussian-smoothed radial distribution around atom i.

Cutoff Integration cutoff, default 5 Å — three or four coordination shells works well.

Broadening σ Gaussian width, default 0.15 Å.

Radial grid points Default 100.

Average over neighbors Smooths s_i over each atom's neighbourhood, sharpening the contrast between ordered and disordered regions.

The result is stored as the per-atom field `local_entropy` and mapped onto the atoms immediately, with a distribution histogram in the dialog. *Lower* (more negative) values mark more ordered, crystal-like environments; higher values mark disordered, liquid-like ones. This makes it an effective order parameter for melting, solid-liquid interfaces and grain boundaries.

11.8 Raman modes

Analysis ⟩ Raman Modes performs a Γ-point factor-group analysis: which optical phonons are Raman-active, IR-active or silent, by symmetry alone.

The method uses no hard-coded character tables. spglib supplies the space group operations of the primitive cell; the character table of the isogonal point group is computed numerically from the class-sum algebra; the mechanical representation $\chi(R) = (\pm 1 + 2\cos\theta)\, N_{\text{unmoved}}(R)$ is reduced into irreducible representations; the acoustic translations are subtracted; and activity follows from the vector (IR) and symmetric polarizability (Raman) representations.

The dialog reports the space group, point group, number of atoms in the primitive cell, and a table of irreps with their degeneracy, optical mode count and activity.

Tip
For silicon in the diamond structure the result is the textbook $\Gamma = T_{2g} + T_{1u}$: one triply degenerate Raman-active optical mode, and the acoustic branch.

Note
Mulliken subscripts are assigned from Cartesian axis geometry. In low-symmetry orthorhombic groups the subscript convention can be permuted relative to a particular textbook's axis choice — the activities and degeneracies are unaffected.

Requires
Needs `spglib` and a periodic structure.

11.9 Volumetric data

Analysis ⟩ Volumetric Data visualises 3D scalar fields: charge densities, electrostatic potentials, wavefunctions, ELF.

11.9.1 Loading fields

Load Field A... and **Load Field B...** read Gaussian `.cube` (including files written by Quantum ESPRESSO's `pp.x`), VASP `CHGCAR` / `LOCPOT` / `PARCHG` / `ELFCAR`, and XCrySDen `.xsf` grids.

Each is reported with its grid dimensions and value range.

Field A defines the geometry; Field B is optional and colours it.

11.9.2 Isosurfaces

The isovalue slider and numeric box re-extract the surface in real time. Extraction uses a marching-cubes-family algorithm on a tetrahedral decomposition of each grid cell, which is topologically unambiguous, with smooth normals from the field gradient and periodic stitching so surfaces close correctly across cell boundaries.

11.9.3 Electrostatic potential maps

Check **Color by Field B (potential map)** to make an EPM: Field A (say the electron density) shapes the surface, and Field B (say the Hartree potential) is interpolated at every surface vertex and mapped through the chosen colormap. The value range is reported so the colour scale is quantitative.

> **Tip**
> The classic recipe: load the charge density as Field A, the local potential as Field B, pick an isovalue on the density that traces the molecular surface, and choose **Coolwarm** — the diverging palette puts the neutral potential at white, so electron-rich and electron-poor regions read immediately.

11.9.4 Slice planes

Slice plane cuts through the volume with a colour-mapped plane: **XY**, **XZ**, **YZ**, or **Custom normal** with an arbitrary direction and an offset slider. Slices can sample either field.

11.9.5 Export

Export Isosurface (OBJ)... writes the triangle mesh with normals for external rendering; **Export Slice (CSV)...** writes the sampled plane as x, y, z, value.

11.10 Adsorption and catalysis

Analysis ⟩ Adsorption & Catalysis finds high-symmetry adsorption sites on a slab and populates them.

11.10.1 Site detection

Opening the dialog detects sites on the top surface automatically and reports a census, for example *4 top, 12 bridge, 4 fcc, 4 hcp*:

`top` Directly above a surface atom.

`bridge` Midway between nearest-neighbour surface atoms.

`fcc`, `hcp` Threefold hollows, distinguished by whether a second-layer atom sits directly underneath (hcp) or not (fcc).

`hollow` A threefold hollow that could not be classified, typically because there is no second layer.

Neighbour vectors are enumerated over periodic images, so the census is correct even for a small

$p(1 \times 1)$ cell where surface atoms bond to their own periodic copies. The table lists every site with its in-plane coordinates and can be filtered by type.

11.10.2 Placing adsorbates

Adsorbate `OH`, `O`, `H`, `CO`, `CHO`, `H2O`, `N2`, `O2`, `NH3`, `CH4`, or any `ase.build.molecule` name or chemical formula you type.

Height Anchor-atom height above the surface layer, default 2 Å.

Place on Selected Sites Adds one copy per selected table row.

Molecules are anchored by the chemically sensible atom and oriented upright: O down for `OH` and `H2O`, C down for `CO` and `CHO`, N down for `NH3`.

11.10.3 Coverage series

The **Coverage series** group generates a systematic set in one step: choose a **Site family** and check the coverages you want — 0.25, 0.50, 0.75, 1.00 ML. Calango places evenly strided subsets of that site family and opens one labelled tab per coverage, ready for a batch of adsorption energy calculations.

11.11 Brillouin zone and k-path builder

Analysis ⟩ Brillouin Zone / k-Path shows the first Brillouin zone as a Wigner-Seitz cell of the reciprocal lattice, with high-symmetry points labelled from ASE's Bravais lattice detection.

Drag to rotate, Shift+drag to pan, wheel to zoom; **Orthographic projection** removes perspective foreshortening when you need to read the zone geometry.

11.11.1 Building a path

Click high-symmetry points in the 3D view to append them to the path. The list shows the sequence with fractional coordinates.

Suggested Loads ASE's suggested path for the lattice.

Break Starts a new discontinuous section, giving paths like $\Gamma \to X \mid M \to R$.

Undo, Clear Remove the last point, or start over.

Points per segment Sampling density, default 40.

11.11.2 Export

Export k-Path... writes the path for an external code:

Format	Typical file
VASP `KPOINTS` (line mode)	`KPOINTS`
Quantum ESPRESSO `K_POINTS crystal_b`	`kpath_qe.in`
CASTEP `SPECTRAL_KPOINT_PATH`	`kpath_castep.cell`
SIESTA `BandLines`	`kpath_siesta.fdf`
ASE / Python script	`kpath_ase.py`

Export Figure (PNG/SVG)... renders the zone at high resolution; the SVG output is vector and drops straight into a paper figure.

CHAPTER 12

Publication Output

12.1 Still images

File ⟩ Import / Export ⟩ Export Image (Ctrl+E) renders the current view off-screen at any resolution, independent of the window size, with 8× multisample anti-aliasing.

Resolution preset 720p, 1080p, 4K, or custom width and height up to 8192 px.

Background **Transparent (PNG only)**, **Solid white**, or a custom colour.

Format PNG or JPEG, chosen by the extension.

> **Tip**
> Transparent PNG is the right choice for figures that will sit on a coloured background in a paper or slide. Combine it with an orthographic, axis aligned view (O, then an alignment button) for a clean, reproducible result.

12.2 Animations

File ⟩ Import / Export ⟩ Export Animation (GIF/MP4) writes either a turntable rotation of a static structure or a playback of a trajectory.

Source **Turntable rotation (360°)** or the current trajectory.

Resolution preset Up to 4K.

Frames per second Playback rate.

Background Including **Transparent (GIF only)**.

Format Animated GIF or H.264 MP4.

> **Requires**
> GIF export needs `pillow`; MP4 needs `imageio` and `imageio-ffmpeg`. MP4 has no alpha channel — asking for a transparent MP4 falls back to a solid background.

12.3 Ray-traced rendering

File ⟩ Import / Export ⟩ Ray-Traced Render exports the active scene — camera, lights, representation, colours, multiple bonds, unit cell — to an external ray tracer for publication-quality output with true shadows and reflections.

Engine **POV-Ray** or **Tachyon**.

Width (px), Height (px) Default 1920 × 1440.

Background **Solid white** or **Viewport color**.

Renderer binary Path to the executable; the bare names `povray` and `tachyon` are resolved on `PATH`. The path is remembered per engine.

Two ways to use it:

Save Scene File... Writes just the scene (`.pov` or `.dat`) for you to render elsewhere, or to edit by hand before rendering.

Render... Writes the scene next to your chosen PNG output and invokes the renderer, streaming its output into the log pane so you can watch progress and diagnose failures.

> **Requires**
> Needs POV-Ray or Tachyon installed separately. If the binary cannot be started, the log says so explicitly rather than failing silently.

12.4 Data exports at a glance

Nearly every analysis tool exports its numbers, so figures can be regenerated in your own plotting stack.

Source	Formats	Typical file
RDF	CSV, DAT	`rdf.csv`
Bond length / angle	CSV, DAT	`bond_lengths.csv`
Structure factor	CSV, DAT	`structure_factor.csv`
XRD pattern	CSV, DAT	`xrd_pattern.csv`
XRD peak list	CSV, DAT	`xrd_peaks.csv`
Warren-Cowley	CSV	`warren_cowley.csv`
Job metric series	CSV, DAT	`energy.csv` ...
Band structure	CSV, DAT	`bands.csv`
PDOS	CSV, DAT	`pdos.csv`
Isosurface mesh	OBJ	`isosurface.obj`
Volumetric slice	CSV	`slice.csv`
Phonon bands / DOS	CSV	`phonon_bands.csv`
k-path	5 codes	Section 11.11
ML datasets	extxyz, ASE db	Section 8.7
Brillouin zone figure	PNG, SVG	`brillouin_zone.svg`

CHAPTER 13

Reference

13.1 Keyboard shortcuts

13.1.1 Viewport

Single-letter shortcuts, no modifier. They yield to text fields while you are typing.

Key	Action
R	Rotation mode
T	Translation (pan) mode
S	Selection mode
I	Insertion mode
D	Distance measurement mode
A	Angle measurement mode
O	Toggle orthographic / perspective
F	Frame structure
Delete / Backspace	Delete selection (Selection mode)

13.1.2 Application

Shortcut	Action
Ctrl+N	New workspace
Ctrl+O	Open structure
Ctrl+T	Open trajectory
Ctrl+Shift+O	Open project
Ctrl+S	Save project
Ctrl+Shift+S	Save structure as
Ctrl+Shift+T	Save trajectory as
Ctrl+E	Export image
Ctrl+W	Close tab
Ctrl+Q	Quit
Ctrl+Z	Undo
Ctrl+Shift+Z	Redo
Ctrl+Shift+A	Add atom
Ctrl+B	Bond editor
Ctrl+R	New calculation
Ctrl+Shift+R	New remote calculation

On macOS, Ctrl in this table is Cmd.

13.1.3 Mouse

Gesture	Action
Left drag	Depends on interaction mode
Middle drag	Pan (any mode)
Shift+left drag	Pan (any mode)
Wheel	Zoom
Left click	Pick atom
Ctrl/Cmd+click	Toggle atom in selection
Double click	Frame structure

13.2 Menu map

File New Workspace · Open (Structure, Trajectory) · Save (Structure As, Trajectory As) · Import / Export (Image, Animation, Ray-Traced Render) · Project Workspace (Open, Save, Save As) · Close Tab · Quit

Edit Undo · Redo · Add Atom · Selection (Change Element, Translate) · Delete Selected Atoms · Bond Editor · Unit Cell (Create Supercell) · Preferences

View Orthographic · Overlays (Show Unit Cell) · Alignment (Frame Structure, XY, XZ, YZ) · Visual Effects · panel toggles

Build Nanomaterial Builder · Surface Slab · Metallic Nanoparticle (Wulff) · Special Quasirandom Structure (SQS) · Normal Modes / Phonon Builder · From Database · Structure Perturbation / Noise

Simulation New Calculation (ASE) · New Remote Calculation · Electronic Bands / PDOS · Dataset Manager (MLIP)

Analysis Structure Factor S(q) · X-Ray Diffraction · Radial Distribution Function · Bond Length / Angle Distributions · Coordination Numbers (CN / GCN) · Warren-Cowley analysis · Local Entropy Analysis · Raman Modes · Volumetric Data · Adsorption & Catalysis · Brillouin Zone / k-Path

Help Documentation · GitHub Repository · About Calango

13.3 Python dependencies by feature

Package	Needed for
`ase`, `numpy`	Everything (structure I/O, building, jobs)
`spglib`	Symmetry detection, Raman modes
`paramiko`	Remote Access panel
`pillow`	GIF animation export
`imageio`, `imageio-ffmpeg`	MP4 animation export
`mace-torch`	MACE machine-learning potentials
`gpaw`	DFT band structures and PDOS
`icet`	Preferred SQS backend (optional)

External binaries: `pw.x` (Quantum ESPRESSO), `siesta`, VASP, `orca`, `povray` or `tachyon`.

13.4 Job directory contents

A typical job directory under `.calango_tmp/` (Section 2.4.1):

File	Written by
`run.py`	The generated (possibly hand-edited) script
`structure.extxyz`	The staged input geometry
`opt.traj`	Geometry optimization trajectory
`md.traj`	Molecular dynamics trajectory
`optimized.extxyz`	Final relaxed geometry
`bands.json`, `pdos.json`	Electronic structure results
`phonon_bands.csv`, `phonon_dos.csv`	Phonon results
`perturbed.extxyz`	Noise ensemble checkpoint
`job.sh`	Scheduler wrapper (remote jobs)
`calango_job.out`, `calango_job.err`	Remote stdout/stderr

13.5 Progress markers

Generated scripts communicate with the interface by printing markers to standard output. If you hand-edit a script or write your own, emitting these keeps the Job panel live:

Marker	Effect
`CALANGO_PROGRESS step total`	Progress bar
`CALANGO_ENERGY step value`	Energy plot (eV)
`CALANGO_TEMP step value`	Temperature plot (K)
`CALANGO_FMAX step value`	Force plot (eV/Å)
`CALANGO_PRESSURE step value`	Pressure plot (GPa)
`CALANGO_TARGET_TEMP value`	Thermostat reference line
`CALANGO_TARGET_PRESSURE value`	Barostat reference line
`CALANGO_CELL` + `CALANGO_FRAME`	Live trajectory streaming
`CALANGO_RESULT ...`	Final summary line

13.6 Troubleshooting

Structures will not open; job features are disabled. ASE is not importable. Run `calango` ↪ `--probe-python` to see which interpreter is being used, then either install ASE there or set `CALANGO_PYTHON` to an environment that has it (Section 1.2).

The space group is `P1` for an obviously symmetric crystal. Numerical noise in the coordinates. Raise **Tolerance** in the **Structure** panel — a relaxed cell often needs 0.01 Å or more.

Bonds are missing or spurious. Adjust **Covalent cutoff multiplier** in the bond editor (Section 6.3), or add and suppress individual bonds by hand. Metallic and ionic systems rarely match a covalent-radius rule.

Double and triple bonds do not appear automatically. By design. Assign them with the **Bond order** buttons (Section 6.4).

A job fails immediately with an import error. The job is running in a different environment from Calango itself. Check the **Execution Environment** status line in the calculation dialog (Section 8.3).

The GPAW backend is greyed out. GPAW is not importable in the selected environment. Point the **Environment** selector at a Conda environment where it is installed.

The remote panel reports a connection failure. Test the same host, port, user and key with `ssh` from a terminal first. Remember that the passphrase field doubles as the key passphrase in **SSH key** mode.

Adsorption site detection finds no bridge or hollow sites. The surface cell is too small to have in-plane neighbours. Build a supercell of the slab first.

The viewport looks soft after enabling depth of field. Expected — the effect trades multisample anti-aliasing for the blur pass (Section 3.7). Disable it before exporting still images.

Animation export fails. Install `pillow` for GIF, or `imageio` and `imageio-ffmpeg` for MP4, in Calango's Python environment.

13.7 Further reading

`README.md` Feature overview and build instructions.

`docs/packaging.pdf` Building the macOS and Linux installers.

`https://github.com/seixas-research/calango` Source, issues and releases.

`https://wiki.fysik.dtu.dk/ase/` ASE documentation — the reference for anything in a generated script.